

7

Genes and inheritance

> 7.1 Chromosomes, genes and DNA

In this topic you will:

- learn that chromosomes contain genes
- find out what genes and chromosomes are made of.

Getting started

Discuss these questions with a partner. Then share your ideas with the rest of the class.

Atoms and cells both contain a nucleus.

- 1 Why do you think these very different structures have the same name?
- 2 Which one is larger?
- 3 What does the nucleus of a cell do?

Key words

chromosomes
DNA
genes

Chromosomes

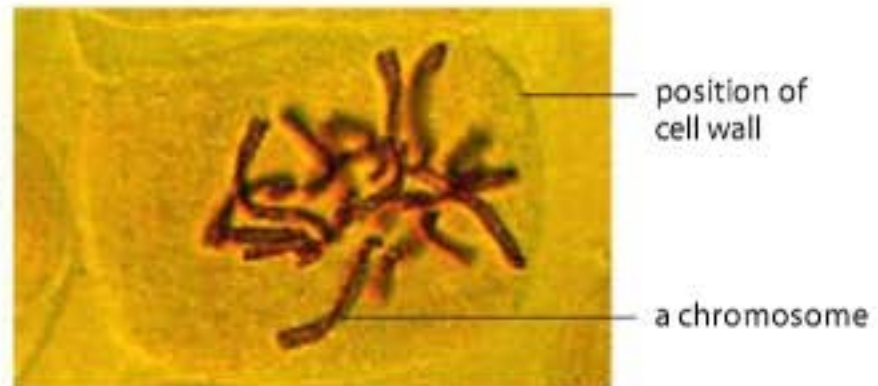
In Stage 7, you learnt that the nucleus of a cell controls its activities. In this unit, you will learn much more about how it does this.

The nucleus of every cell contains threads called **chromosomes**.

Chromosomes were discovered in the nineteenth century. This was when scientists first had sufficiently powerful microscopes, and were first able to see very small things, such as the structure of cells.

When you looked at cells using a microscope, you almost certainly did not see any chromosomes. This is because chromosomes only become visible with a light microscope when a cell is dividing. Chromosomes also need to be coloured using special stains, in order to be able to see them.

This photograph shows a plant cell that is just beginning to divide into two cells. The cells have been stained with a special dye that colours the chromosomes dark red. You cannot really see any of the other structures in the cell.



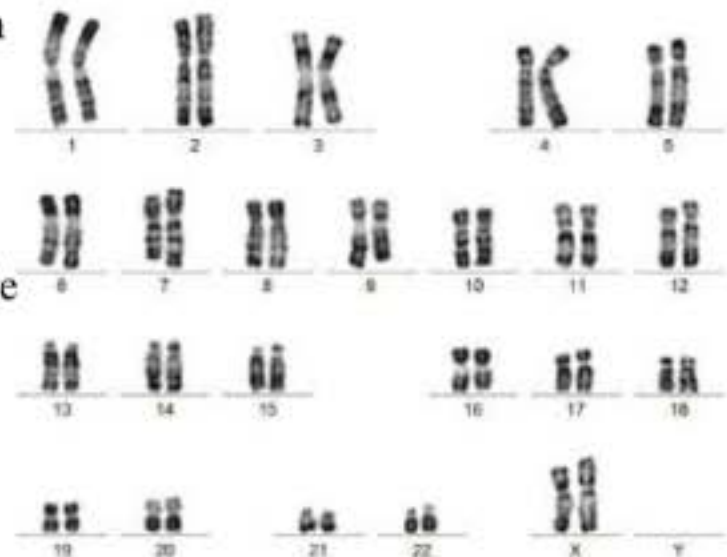
This photograph shows a chromosome from a fruit fly. Fruit flies have especially thick chromosomes, called giant chromosomes, which are much easier to see than most chromosomes. You can see that this chromosome is a long, striped thread.



Different species of organism have different numbers of chromosomes in the nuclei of their cells. Human cells have 46 chromosomes. Fruit fly cells have eight chromosomes. Mango trees have 40 chromosomes.

Scientists number the chromosomes in a cell according to how long the chromosomes are. In a human cell, the longest chromosome is chromosome 1, the next longest is chromosome 2, and so on.

We have two of each kind of chromosome. This photograph shows all the chromosomes in a human cell. It has been made by cutting and pasting pictures of the individual chromosomes. They have then been lined up in size order. They do not really line themselves up like this!



If you look very closely at this photograph, you may be able to see that each chromosome looks like a cross shape. You can see this really clearly in the photograph at the start of the next topic. This is because – just before a cell divides – each chromosome makes a copy of itself. The two copies stay joined together part-way along their length.

Activity 7.1.1**Making models of chromosomes in a cell**

Work in a group of two or three for this activity.

You are going to make a set of chromosomes in a cell of a chosen species of animal.

You can use whatever materials you like to make the models of chromosomes.

For example, you could use string or wool. You will also need a large sheet of paper.

First, use the internet to find out how many chromosomes your species has in its cells.

Next, draw an outline of a cell on a large sheet of paper. Do not make it a perfect circle – draw it freehand. Draw a large nucleus inside the cell.

Now make your chromosomes. Remember that there are two chromosomes of each type. So, if your animal has 32 chromosomes, you need to make two sets of 16.

Stick the chromosomes inside the nucleus of the cell.

Write a heading on the sheet of paper to say which species of animal your cell represents.

Self-assessment

How satisfied are you with your model of a cell containing chromosomes?

How does it compare with the models that other groups have made?

What would you do to improve your model, if you had different materials to use, and more time?

Questions

- 1 Explain why we cannot usually see the chromosomes in a cell, even using a microscope.
- 2 Red blood cells do not contain a nucleus. Do you think they contain any chromosomes? Explain your answer.

Genes

Each chromosome is made up of hundreds of different **genes**. The genes are arranged in a particular sequence along the chromosome. Each gene helps to control a particular characteristic in the organism. On this diagram of a short chromosome, the stripes represent different genes.



Scientists are still finding out which genes are found on each type of chromosome in humans. On chromosome 1, for example, we know that there are about 2000 different genes. Chromosome 15 is a much shorter chromosome, and it has about 600 different genes.

Discovering exactly what each of these genes do is not easy, but scientists are learning more all the time. For example, we know that two genes on chromosome 15 help to determine eye colour. Everyone has genes for eye colour in the same place on their chromosome 15s. But there are different versions of these genes, so one person could have a chromosome 15 with eye colour genes that give them blue eyes, and another could have a chromosome 15 with eye colour genes that give them brown eyes. That is why the two sisters in this photograph have different eye colours.



DNA

Chromosomes are made of a chemical substance called **DNA**. Each chromosome is one enormously long molecule of DNA.

This means that genes are also made of DNA.

A DNA molecule has a shape like a twisted ladder. This shape is called a double helix. One gene could be a length of DNA with about 2500 of these twists. We cannot see these twists when we use microscopes to look at chromosomes because DNA molecules are much too small to see.

DNA was first discovered in the 1950s. Since then, scientists have found out a great deal about how the DNA in genes helps to determine the characteristics of humans and other organisms. The DNA in a cell determines what the cell does. It contains a complete set of instructions to make a functioning cell, and a whole organism. If you continue to study science to IGCSE or O level, you will find out much more about this.



Questions

- 3 Explain the difference between a chromosome and a gene.
- 4 All fruit flies have a gene for wing shape, in the same place on their chromosome 2. But some fruit flies have normal wings, and some have very tiny wings – so small that they cannot fly. What can explain this?

Summary checklist

- ☐ I can explain that chromosomes are made of DNA.
- ☐ I can explain that each chromosome contains many genes.
- ☐ I can explain that genes help to determine an organism's characteristics.